

**"THE GUARDIAN": A MODULAR AI FOR
PROACTIVE HUMAN-LAYER CYBERSECURITY ON
SOCIAL MEDIA.**

Project ID: 25-26J-041

Project Proposal Report

Herath, H. M. G. D. K. (IT22361486)

Supervisor – Kavinga Yapa Abeywardena

Co-supervisor – Deemantha Siriwardana

B. Sc. (Hons) Degree in Information Technology (Cyber Security)

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

**"THE GUARDIAN": A MODULAR AI FOR
PROACTIVE HUMAN-LAYER CYBERSECURITY ON
SOCIAL MEDIA.**

Project ID: 25-26J-041

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology (Cyber Security)

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Herath, H. M. G. D. K.	IT22361486	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor:



Date: 29/08/2025



Signature of the Co-Supervisor

Date: 30/08/2025

Abstract

The increasing prevalence of human-layer cybersecurity threats on social media platforms exposes users to psychological manipulation, misinformation, and the leakage of personally identifiable information (PII). Existing reactive moderation systems are ill-equipped to provide the real-time, user-centric defense necessary in this landscape.

This research proposes "The Guardian", a modular, AI-powered browser extension designed to address this gap. The system integrates specialized Natural Language Processing (NLP) models with a FastAPI backend to proactively identify linguistic threats. This proposal details the development of a core module focused on detecting PII exposure and contextual operational security (OpSec) risks using a hybrid approach of Named Entity Recognition (NER) and rule-based pattern matching. By providing non-intrusive, context-aware alerts, the system empowers users to make informed security decisions without imposing censorship. This work's novelty lies in its proactive, educational stance and its specific focus on combining PII detection with OpSec risk analysis for the unique challenges of social media. The research will evaluate the module's detection accuracy, performance, and usability to validate its effectiveness in enhancing individual cybersecurity posture.

Keywords: Cybersecurity, Natural Language Processing, Human-Layer Security, PII, OpSec.

TABLE OF CONTENTS

Declaration	i
Abstract	ii
List of Figures	v
List of Tables.....	vi
List of Abbreviations.....	vii
1. Introduction	1
2. Background & Literature Survey	3
2.1 The Human Factor and Social Media as a Threat Vector	3
2.2 Automated Detection of Personally Identifiable Information (PII)	3
2.3 Core Technologies and Methodological Foundations	4
2.4 The OpSec Gap and the Need for User-Centric Design	5
3. Research Gap	6
4. Research Problem.....	7
5. Objectives.....	8
5.1 Main Objective.....	8
5.2 Specific Objectives.....	8
6. Methodology	10
6.1 Overall System Architecture	10
6.2 PII & OpSec Module: Design and Implementation	11
6.3 Data Collection and Preparation	14
6.4 Technical Requirements & Resources	14
6.5 Evaluation Plan	15
7. Description of Personnel and Facilities.....	16
7.2 Facilities and Resources	17
8. Project requirements.....	18

8.1 User Requirements	18
8.2 Functional Requirements (FR)	18
8.3 Non-Functional Requirements (NFR)	19
8.4 Use Cases	20
8.5 Test Cases	22
8.6 Wireframes	23
9. Project Plan and Work Breakdown Structure (WBS)	24
Phase 1: Research and Foundation (Duration: 4 Weeks)	24
Phase 2: Core Module Development (Duration: 6 Weeks - Parallel Work)	25
Phase 3: System Integration and Frontend Development (Duration: 4 Weeks)	26
Phase 4: Testing and Evaluation (Duration: 3 Weeks)	26
Phase 5: Finalization and Reporting (Duration: 4 Weeks)	27
10. Budget and Budget Justification	28
11. Commercialization and Entrepreneurship Potential	29
11.1 Market Opportunity	29
11.2 Commercialization Pathways	29
11.3 Value Proposition	30
11.4 Entrepreneurial Impact	30
References	31
Appendices	33
Appendix A: Project Gantt Chart	33
Appendix B: UI Wireframes	34

LIST OF FIGURES

Figure Number	Figure Title	Page
Figure 6.1	High-level system architecture of "The Guardian," showing the data flow from the client-side browser extension to the backend analysis modules and back.	11
Figure 6.2	Internal data processing pipeline of the PII & OpSec module, illustrating the parallel PII detection and OpSec risk analysis paths.	12

LIST OF TABLES

Table Number	Table Title	Page
Table 8.1	Functional Requirements (FR) for the PII & OpSec Module	18
Table 8.2	Non-Functional Requirements (NFR) for the PII & OpSec Module	19
Table 8.3	Tentative Test Cases for System Validation	22
Table 9.1	Phase 1 of the Work Breakdown Structure: Research and Foundation	24
Table 9.2	Phase 2 of the Work Breakdown Structure: Core Module Development	25
Table 9.3	Phase 3 of the Work Breakdown Structure: System Integration and Frontend Development	26
Table 9.4	Phase 4 of the Work Breakdown Structure: Testing and Evaluation	26
Table 9.5	Phase 5 of the Work Breakdown Structure: Finalization and Reporting	27

LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
B2B	Business-to-Business
B2C	Business-to-Consumer
BERT	Bidirectional Encoder Representations from Transformers
FR	Functional Requirement
HCI	Human-Computer Interaction
LMS	Learning Management System
NER	Named Entity Recognition
NFR	Non-Functional Requirement
NIC	National Identity Card
NLP	Natural Language Processing
OpSec	Operational Security
OSINT	Open-Source Intelligence
PII	Personally Identifiable Information
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
UX	User Experience
WBS	Work Breakdown Structure

1. INTRODUCTION

The paradigm of cybersecurity has fundamentally evolved from protecting network perimeters to securing the most unpredictable element: the human user. Social media platforms, while fostering global connectivity, have simultaneously emerged as the primary vector for a new class of "human-layer" threats that exploit psychology, trust, and cognitive biases [1]. These threats range from coordinated misinformation campaigns to sophisticated social engineering scams. A particularly insidious and often underestimated risk is the unintentional disclosure of Personally Identifiable Information (PII) and contextual data, which creates significant Operational Security (OpSec) vulnerabilities [2].

This problem is exacerbated by the well-documented "privacy paradox," a cognitive dissonance where users express high concern for their digital privacy yet continue to engage in high levels of self-disclosure online [3]. This behavior provides a rich source of data for malicious actors who employ Open-Source Intelligence (OSINT) techniques to harvest this information for nefarious purposes, including identity theft, targeted phishing, and even physical harm [4]. The malicious practice of "doxing", publishing an individual's PII without consent, has become a severe form of online harassment, with research demonstrating the feasibility of automatically detecting such leaks on platforms like Twitter [5]. While these studies focus on malicious third-party actions, a critical gap exists in addressing unintentional self-disclosure, where users inadvertently become the source of their own vulnerability.

Current platform-native moderation systems are largely reactive, acting only after harm has occurred, and are not designed to provide preventative, personalized guidance. This research proposes the development of a specialized module within the "The Guardian" system, a proactive, AI-powered browser extension. This specific module is designed to address the challenge of PII and OpSec risk by providing users with real-time, context-aware alerts. By leveraging a hybrid approach of state-of-the-art Natural Language Processing (NLP) for PII detection and a rule-based engine for identifying risky contextual patterns, the module aims not to censor, but to educate. It acts as a cognitive "nudge," helping users align their online actions with their privacy

intentions, thereby fortifying their personal security posture in an increasingly complex digital world.

2. BACKGROUND & LITERATURE SURVEY

This section critically reviews the existing body of knowledge pertinent to the detection of PII and OpSec risks on social media. It examines the landscape of human-layer threats, evaluates current technological approaches for data protection, and identifies the specific gaps in research that this project aims to address.

2.1 The Human Factor and Social Media as a Threat Vector

The contemporary cybersecurity landscape is increasingly defined by threats that target the human element. Ghafir et al. [1] establish the "human factor" as a primary vulnerability, arguing that technical security measures are frequently bypassed through the exploitation of human psychology and error. Social media platforms, by their very nature, amplify this vulnerability. They are designed to encourage sharing and social interaction, which inadvertently creates an environment ripe for data leakage and manipulation [2]. This is compounded by the "privacy paradox," a documented psychological phenomenon where individuals' stated privacy concerns often do not align with their actual disclosure behaviors, leading them to share more information than is prudent [3].

The consequence of this oversharing is the creation of a vast repository of public data that can be systematically collected and analyzed using Open-Source Intelligence (OSINT) techniques [4]. Malicious actors leverage OSINT to build detailed profiles of potential victims for highly targeted attacks, including social engineering, identity theft, and ransomware campaigns that leverage personal details to increase their credibility [6]. The proactive identification of sensitive data at the point of disclosure is therefore a critical, yet underdeveloped, area of personal cyber defense.

2.2 Automated Detection of Personally Identifiable Information (PII)

The automated detection of PII from unstructured text is a well-established discipline, often termed "de-identification" in formal domains like clinical medicine. Research in this area, such as the shared tasks on de-identifying psychiatric intake records by Stubbs and Uzuner [7], provides a strong foundation for the technical challenges involved. These studies demonstrate the effectiveness of NLP models in

handling complex, narrative-style text, which is analogous to the informal language used on social media.

More recently, research has shifted to the specific context of social media platforms. The work by Karimi et al. [5] on the automated detection of "doxing" on Twitter is particularly relevant. Their successful hybrid approach, which combined pattern matching with machine learning, validates the methodological direction of this project. However, their focus remains on the detection of malicious, third-party disclosures. Furthermore, academic efforts to create standardized benchmarks for sensitive data detection, such as the SPeDaC benchmark [8], have advanced the capabilities of Transformer models but also highlight the need for domain-specific evaluation. There remains a clear research gap in developing tools focused on preventing *unintentional self-disclosure* and educating users about their personal security posture.

2.3 Core Technologies and Methodological Foundations

The primary technology for identifying contextual PII is Named Entity Recognition (NER), a subfield of NLP. Modern NER systems have moved beyond simple dictionary lookups to employ sophisticated deep learning models. Transformer architectures, exemplified by BERT (Bidirectional Encoder Representations from Transformers) [9], represent the current state of the art. Their key innovation is the ability to understand the context of a word, which is critical for accurately identifying entities in ambiguous social media text. Recent surveys of the field confirm the superior performance of these models over older techniques and provide a strong justification for their selection in this project [10].

To train and effectively evaluate such models, specialized, context-rich datasets are required. The development of synthetic datasets like PANORAMA [11], which is specifically designed with PII embedded in realistic social media scenarios, underscores the importance of domain-specific data. While a purely model-based approach is powerful, a hybrid methodology is often superior. For structured PII with predictable formats (e.g., phone numbers, email addresses), rule-based methods using Regular Expressions (Regex) offer unparalleled precision and computational

efficiency. This project will therefore adopt a hybrid approach, combining the contextual power of a Transformer-based NER model with the precision of Regex rules.

2.4 The OpSec Gap and the Need for User-Centric Design

While detecting explicit PII is a solved problem to a large extent, a significant gap exists in the automated detection of contextual Operational Security (OpSec) risks. An OpSec risk is not about the data itself, but what the data *reveals* (e.g., a post about an upcoming vacation implies an empty home). This level of contextual inference is largely absent from existing academic research and commercial tools.

Furthermore, the usability of security tools is paramount to their effectiveness. The phenomenon of "security fatigue" occurs when users are inundated with excessive, disruptive warnings, leading them to ignore security advice altogether [12]. This finding strongly motivates the design of a non-intrusive user interface. The proposed solution's use of subtle, on-demand alerts is a conscious design choice based on principles of usable security, aiming to provide effective "nudges" that encourage safer behavior without contributing to user fatigue.

3. RESEARCH GAP

The comprehensive literature survey reveals several critical gaps in the existing research and available tools for protecting users from human-layer threats on social media. While significant progress has been made in specific areas, the following distinct gaps persist:

- **Focus on Malicious Intent over User Error:** Current state-of-the-art research, such as the work on doxing detection [5], is primarily focused on identifying malicious third-party attacks. There is a notable lack of tools and research dedicated to addressing unintentional, self-inflicted privacy risks where users themselves are the source of the data leakage.
- **Absence of Automated OpSec Analysis:** The literature extensively covers the detection of explicit PII (e.g., names, ID numbers) [7], [8]. However, there is a significant gap in the automated analysis of contextual Operational Security (OpSec) risks. No existing research presents a methodology for automatically inferring risks from the combination of non-sensitive keywords and temporal information (e.g., identifying future travel plans or routines).
- **Lack of Integrated, User-Centric Educational Tools:** Existing solutions are fragmented. They are either backend moderation systems designed for platforms or academic models focused on a single task. There is a clear need for an integrated, user-facing system that combines multiple detection capabilities and presents them not as a censorship mechanism, but as a real-time educational tool designed to improve user security literacy.
- **Gap in Usability-Focused Security Design:** While the psychological drivers of oversharing [3] and the pitfalls of poor security UI design ("security fatigue") [12] are understood, few security tools have explicitly incorporated these principles into their core design. A gap exists for a tool designed from the ground up to be a non-intrusive, psychologically-informed security "nudge."

4. RESEARCH PROBLEM

Based on the identified research gaps, this project addresses a central research problem that can be articulated as follows:

How can a hybrid model, combining state-of-the-art Natural Language Processing for Named Entity Recognition with a rule-based engine for contextual analysis, be designed and implemented to effectively provide **real-time, proactive, and educational alerts** for both explicit Personally Identifiable Information (PII) disclosures and implicit Operational Security (OpSec) risks within the unique linguistic environment of social media?

This overarching problem encompasses several sub-problems that this research will investigate:

1. How to accurately adapt and fine-tune pre-trained Transformer models to recognize PII within informal, noisy social media text?
2. How to design an effective set of rules to identify contextual OpSec risks with high precision while minimizing false positives?
3. How to integrate these two distinct detection methods into a single, efficient module that can operate in near real-time?
4. How to present the output of this module to the end-user in a manner that is informative and effective without inducing security fatigue?

5. OBJECTIVES

This research aims to address the identified research problem through the achievement of one main objective, which is further broken down into four specific, measurable sub-objectives.

5.1 Main Objective

To design, implement, and evaluate a robust, real-time module for the "The Guardian" system that accurately detects and provides educational alerts for both explicit Personally Identifiable Information (PII) disclosures and implicit Operational Security (OpSec) risks in user-generated text on social media.

5.2 Specific Objectives

To develop a high-precision PII detection engine by [End of Month 4]. This will be achieved by creating a hybrid model that combines a fine-tuned, Transformer-based Named Entity Recognition (NER) component for contextual entities (e.g., names, locations) with a rule-based Regular Expression (Regex) component for structured entities (e.g., phone numbers, email addresses). The success of this objective will be measured by achieving an F1-score of at least 0.90 on a custom-curated, annotated test dataset of social media posts.

To design and implement a novel OpSec risk analysis engine by [End of Month 5]. This involves identifying a minimum of five distinct, high-risk OpSec patterns (e.g., future travel, financial distress, routine disclosure) and implementing a rule-based system to detect these patterns by analyzing the semantic context surrounding any identified PII. The engine's effectiveness will be measured by its ability to correctly classify at least 95% of test cases in a targeted OpSec scenario dataset.

To integrate the PII and OpSec engines into a single, efficient API endpoint by [End of Month 6]. This objective will be accomplished by wrapping the developed models into a cohesive Python module and exposing its functionality via a FastAPI endpoint. The endpoint must accept raw text as input and return a structured JSON object detailing all detected risks. The performance will be measured by ensuring the

average processing and response time for a typical social media post (under 280 characters) is less than 500 milliseconds.

To conduct a comprehensive performance and usability evaluation of the completed module by [End of Month 7]. This involves a quantitative analysis of the module's precision, recall, and F1-score on the final blind test set, and a qualitative assessment of the clarity and usefulness of the generated alerts. The final evaluation will be documented in a detailed report comparing the performance of different model architectures (e.g., BERT vs. DistilBERT) on the key metrics.

6. METHODOLOGY

This section outlines the systematic approach that will be followed to achieve the project's objectives. It details the overall system architecture of "The Guardian," the specific internal workflow of the PII/OpSec module, the data collection and preparation strategy, and the plan for rigorous evaluation.

6.1 Overall System Architecture

The proposed "Guardian" system is designed as a client-server application, comprising a lightweight browser extension (the client) and a powerful backend analysis engine (the server). This architecture is chosen to balance real-time responsiveness with intensive computational processing, while ensuring user privacy and a non-intrusive experience. The workflow, as depicted in Figure 1, is as follows:

1. **Text Extraction:** The browser extension, running on the user's machine, identifies and extracts text content from social media posts and comment fields as the user browses.
2. **Secure API Request:** The extracted text is securely transmitted via an HTTPS request to the backend API. No user metadata is sent, only the raw text for analysis.
3. **Backend Processing:** The FastAPI backend receives the request and acts as an orchestrator, concurrently passing the text to the four specialized analysis modules.
4. **Modular Analysis:** Each module (PII/OpSec, Hate Speech, Misinformation, Scam) processes the text and returns a structured JSON response indicating any detected threats.
5. **Aggregated Response:** The backend aggregates the responses from all modules into a single JSON object and returns it to the browser extension.
6. **User Notification:** The extension parses the JSON response. If a threat is detected by any module, it displays a subtle, context-aware icon next to the corresponding text. Hovering over this icon reveals a tooltip with a clear,

educational explanation of the potential risk, a design choice informed by research on avoiding "security fatigue" [12].

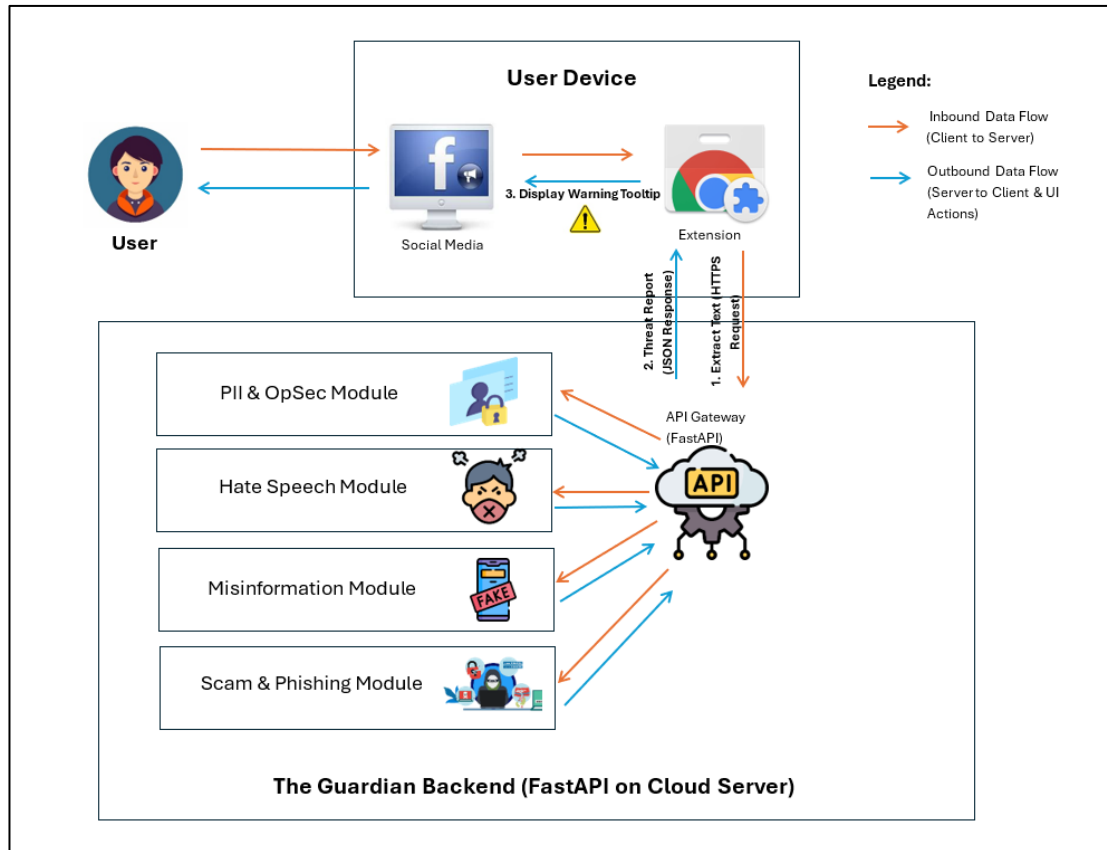


Figure 6.1: High-level system architecture of "The Guardian," showing the data flow from the client-side browser extension to the backend analysis modules and back.

6.2 PII & OpSec Module: Design and Implementation

The PII & OpSec module is the core component of this research and will be developed using a hybrid approach to maximize both accuracy and coverage. The internal workflow, illustrated in Figure 2, is designed as a two-stage pipeline.

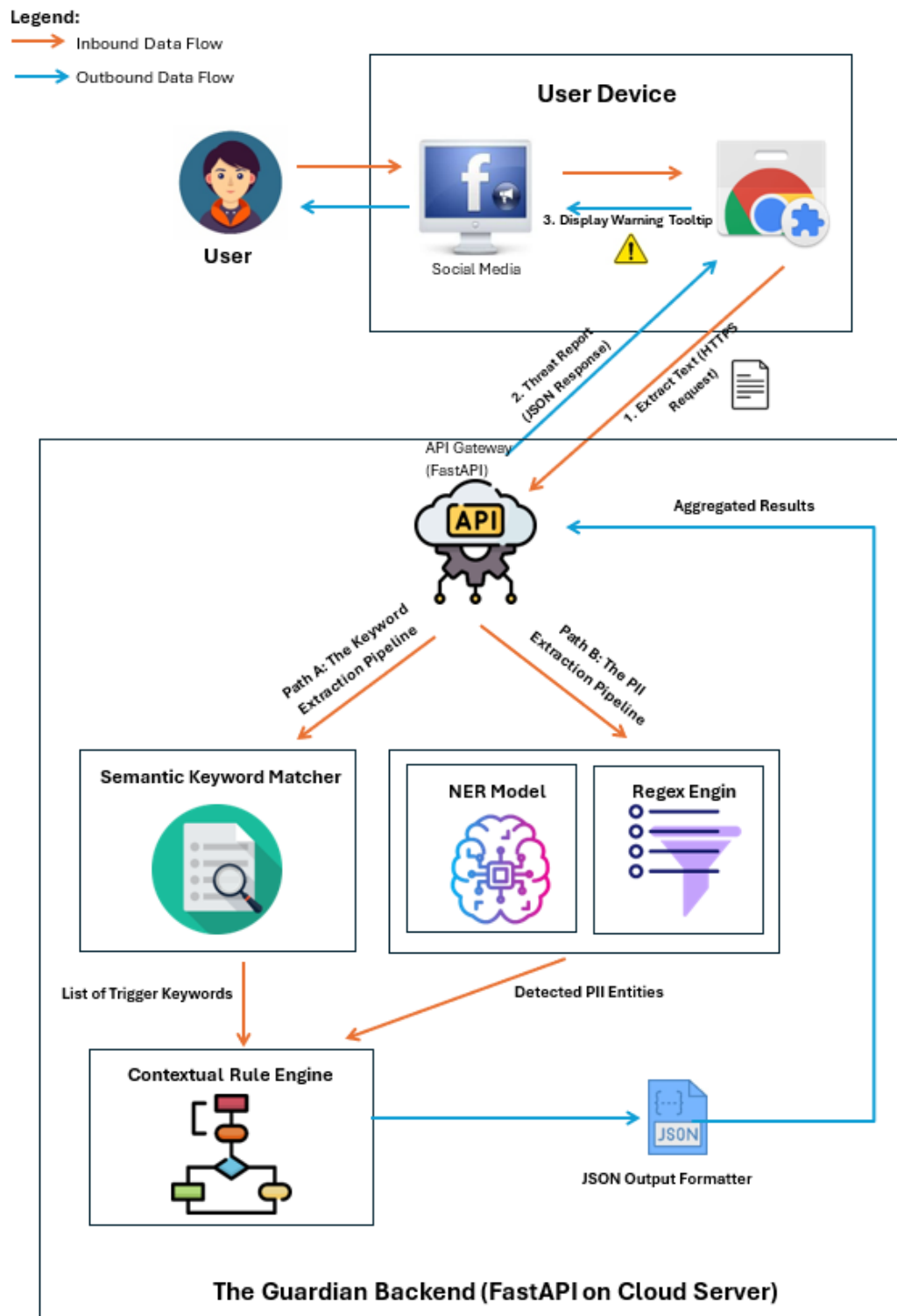


Figure 6.2: Internal data processing pipeline of the PII & OpSec module, illustrating the parallel PII detection and OpSec risk analysis paths.

Stage 1: PII Detection (Parallel Processing)

This stage identifies explicit instances of Personally Identifiable Information using two parallel methods:

- **Named Entity Recognition (NER):** A pre-trained Transformer model (e.g., BERT [9]) will be fine-tuned to identify contextual and unstructured PII such as names of individuals and specific locations. This model-based approach is necessary for entities that do not follow a fixed pattern.
- **Rule-Based Matching (Regex):** A set of highly precise Regular Expressions will be developed to detect structured PII. This is the most efficient method for entities with predictable formats, such as Sri Lankan phone numbers, National Identity Card (NIC) numbers, and email addresses.

The outputs from both methods are combined into a comprehensive list of all PII entities found in the text.

Stage 2: OpSec Risk Analysis (Contextual Rule Engine)

This stage analyzes the text for contextual risks. It uses the output from Stage 1 as a key input.

- **Semantic Keyword Triggering:** The system first scans the text for a predefined dictionary of keywords associated with OpSec risks (e.g., "trip," "vacation," "home alone," "lost my job," "need money").
- **Contextual Rule Engine:** If both a PII entity (from Stage 1) and an OpSec keyword are found within the same text, a rule engine is triggered. This engine applies a set of logical rules to determine the specific type of OpSec risk. For example:
 - IF (ENTITY_TYPE == LOCATION) AND (KEYWORD IN ["trip", "flight", "vacation"]) AND (TEMPORAL_KEYWORD IN ["next week", "tomorrow"]) THEN RISK = TRAVEL_EXPOSURE
 - IF (PII_ENTITY_COUNT > 2) AND (KEYWORD IN ["help", "urgent", "need"]) THEN RISK = OVERSHARING_DISTRESS_SIGNAL

6.3 Data Collection and Preparation

To train and rigorously evaluate the module, a custom dataset will be developed, following the principles of recent synthetic PII dataset creation methodologies like PANORAMA [11].

1. **Corpus Acquisition:** A base corpus of anonymized, public social media posts (e.g., from Twitter) will be collected.
2. **Data Synthesis & Annotation:** Synthetic but realistic PII (e.g., generated Sri Lankan names, phone numbers) and OpSec scenarios will be programmatically inserted into the base corpus.
3. **Dataset Partitioning:** The final annotated dataset will be split into three sets: Training (70%), Validation (15%), and a final blind Test set (15%) to ensure unbiased evaluation of the model's performance.

6.4 Technical Requirements & Resources

The successful execution of this project relies on open-source software and standard academic computing resources. No specialized hardware or budget is required. The necessary materials include:

- **Software & Libraries:** Python (version 3.8+) will serve as the primary programming language. Key libraries will include Hugging Face Transformers for accessing pre-trained NLP models, spaCy for supplementary NLP tasks, FastAPI for API development, and PyTorch or TensorFlow as the deep learning backend.
- **Development Environment:** A standard development computer with sufficient RAM (16GB recommended) for model fine-tuning. Version control will be managed using Git and GitHub.
- **Data Sources:** Publicly available text corpora (e.g., from academic datasets or Twitter APIs, subject to terms of service) will be used as the base for creating the evaluation dataset.

6.5 Evaluation Plan

The success of the module will be evaluated against the specific objectives using a multi-faceted approach:

- **PII Detection Performance:** The accuracy of the PII detection engine will be measured using standard classification metrics:
 - **Precision:** The proportion of identified entities that were correct.
 - **Recall:** The proportion of all actual entities that were correctly identified.
 - **F1-Score:** The harmonic mean of Precision and Recall, providing a single measure of overall accuracy.
- **OpSec Engine Accuracy:** The rule-based OpSec engine will be evaluated based on its classification accuracy on a curated set of scenario-based test cases.
- **Performance Benchmarking:** The module's latency will be measured by calculating the average end-to-end processing time (from text input to JSON output). This will be benchmarked for different model sizes, specifically comparing a BERT-base model with DistilBERT. The original DistilBERT paper demonstrates that it retains 97% of BERT's language understanding capabilities while being 60% faster and 40% smaller, making it a prime candidate for a latency-sensitive application like this one [13].
- **Qualitative Assessment:** The clarity, usefulness, and perceived intrusiveness of the generated user alerts will be qualitatively assessed through a small-scale user feedback survey.

7. DESCRIPTION OF PERSONNEL AND FACILITIES

This section details the composition of the project team, the specific roles and responsibilities of each member, and the facilities and resources available for the successful execution of this research.

7.1 Personnel

The project team comprises four final-year B. Sc. (Hons) in Information Technology (Cyber Security) undergraduate students, each specializing in areas relevant to the project's success, including cybersecurity, software development, and artificial intelligence. The roles are clearly defined to ensure accountability and comprehensive coverage of all project tasks.

- **M1 - Herath, H. M. G. D. K. (IT22361486) - Project Lead & PII/OpSec Module Lead:**
 - Responsible for the overall project coordination, timeline management, and ensuring cohesive integration of all modules.
 - Leads the end-to-end development of the PII & OpSec risk detection module, including data synthesis, model implementation, and performance evaluation.
- **M2 - De Alwis, H. P. G. H. S. (IT22317926) - Misinformation Module Lead:**
 - Responsible for researching propaganda techniques and developing the specialized AI module for detecting manipulative language.
- **M3 - Fernando, M. R. I. (IT22366054) - Hate Speech Module Lead:**
 - Responsible for researching hate speech linguistics and developing the specialized AI module for the detection of toxic and harmful language.
- **M4 - Vishwa J. R. (IT22916112) - Scam & Phishing Module Lead & UI/UX Lead:**

- Responsible for researching online scam tactics and developing the specialized AI module for detecting social engineering attempts.
- Leads the design and development of the browser extension's user interface, including the creation of wireframes and user feedback collection.

7.2 Facilities and Resources

The project will be executed using standard, readily available academic and open-source resources. No specialized or high-cost facilities are required.

- **Laboratory Facilities:** The team will have full access to the computing laboratories at the Sri Lanka Institute of Information Technology (SLIIT), which provide the necessary networking and hardware infrastructure for development and testing.
- **Hardware:** All development and model training will be conducted on personal laptops and university-provided desktop computers, which meet the recommended specifications (e.g., 16GB RAM) for model fine-tuning.
- **Software:** The project will exclusively use open-source software, including:
 - **Programming Language:** Python (v3.8+)
 - **Core Libraries:** Hugging Face Transformers, PyTorch/TensorFlow, spaCy, FastAPI.
 - **Development Tools:** Visual Studio Code, Git, and GitHub for version control and collaboration.
- **Data Sources:** Datasets for model training and evaluation will be sourced from publicly available academic repositories (e.g., Kaggle, Papers with Code) and through the creation of a custom synthetic dataset. The team will have access to the SLIIT library's digital resources for literature review and research.

8. PROJECT REQUIREMENTS

This section defines the specific functional and non-functional requirements for the PII & OpSec risk detection module. It also outlines the system's intended users and provides tentative use cases and test cases that will guide the development and validation process.

8.1 User Requirements

The primary user of this module is any individual who uses social media and is concerned about their digital privacy and security, but may lack the technical expertise to constantly self-monitor their online disclosures. The user requires a tool that:

- Operates automatically in the background without requiring manual intervention.
- Provides simple, understandable, and actionable feedback on potential privacy risks.
- Does not censor, block, or store their personal data.
- Enhances their awareness of PII and OpSec risks over time.

8.2 Functional Requirements (FR)

Functional requirements specify the concrete actions the system must be able to perform.

Table 8.1: Functional Requirements (FR) for the PII & OpSec Module

ID	Requirement	Priority
FR1	The system shall scan user-generated text within designated social media fields in real-time.	High
FR2	The system shall detect and classify structured PII, including phone numbers (Sri Lankan format), National Identity Card (NIC) numbers, and email addresses.	High
FR3	The system shall detect and classify unstructured PII, including personal names and specific locations (e.g., city names, building names).	High

FR4	The system shall analyze the text for predefined contextual OpSec risk patterns (e.g., future travel plans, disclosure of financial distress).	High
FR5	The system shall generate a structured JSON output detailing all detected PII entities and OpSec risks, including their type and location within the text.	High
FR6	The system shall provide a visual, non-intrusive alert on the user interface adjacent to the text containing a detected risk.	Medium
FR7	The system shall , upon user interaction (e.g., hover), display a tooltip containing a concise, educational explanation of the detected risk.	Medium

8.3 Non-Functional Requirements (NFR)

Non-functional requirements define the quality attributes and constraints of the system.

Table 8.2: Non-Functional Requirements (NFR) for the PII & OpSec Module

ID	Requirement	Metric
NFR1	Performance: The end-to-end analysis (text input to JSON output) for a standard post (<280 characters) shall complete in under 500 milliseconds.	Average response time
NFR2	Accuracy: The PII detection model shall achieve a minimum F1-score of 0.90 on the blind test set.	F1-Score
NFR3	Privacy: The backend service shall not log or store any user-submitted text content post-analysis.	Code review, System design
NFR4	Usability: The user interface alerts shall not obscure the underlying social media content or disrupt the standard user workflow.	Heuristic evaluation, User feedback
NFR5	Security: All communication between the browser extension and the backend API shall be encrypted using HTTPS.	Network traffic analysis
NFR6	Modularity: The PII/OpSec module shall be implemented as a self-contained component with a clearly defined API interface to allow for independent testing and integration.	Code structure review

8.4 Use Cases

Use Case 1: Detection of Unintentional PII Leakage

- **Actor:** Social Media User
- **Precondition:** User is logged into a social media platform and composing a new post or comment.
- **Scenario:**
 1. User types: "Great meeting you all! If you have any questions, just email me at mytest.email@example.com."
 2. As the user types, the "Guardian" browser extension sends the text to the backend.
 3. The PII/OpSec module's Regex component identifies "mytest.email@example.com" as an entity of type EMAIL.
 4. The module returns a JSON response indicating the PII detection.
 5. A subtle shield icon appears next to the text box.
 6. User hovers over the icon and a tooltip appears: "Potential PII Risk: You may be publicly sharing an email address."
- **Postcondition:** The user is made aware of the PII disclosure before posting.

Use Case 2: Detection of OpSec Risk (Future Travel)

- **Actor:** Social Media User
- **Precondition:** User is logged into a social media platform.
- **Scenario:**
 1. User types: "Counting down the days! Our flight to London is next Tuesday. So excited!"
 2. The extension sends the text to the backend.

3. The PII/OpSec module's NER component identifies "London" as LOCATION. The OpSec engine identifies the keywords "flight" and "next Tuesday."
 4. The Contextual Rule Engine triggers the TRAVEL_EXPOSURE risk rule.
 5. The module returns a JSON response indicating the OpSec risk.
 6. An alert icon appears. The user hovers and sees the tooltip: "Potential OpSec Risk: Sharing future travel plans can signal that your home may be unattended."
- **Postcondition:** The user is educated about the potential security implications of their post.

8.5 Test Cases

The following test cases represent a sample of the scenarios that will be used to validate the module's functional requirements.

Table 8.3: Tentative Test Cases for System Validation

Test Case ID	Requirement	Test Description	Expected Result
TC-PII-01	FR2	Input text containing a valid Sri Lankan phone number: "call me on 0771234567".	The system correctly identifies "0771234567" as a PHONE_NUMBER entity.
TC-PII-02	FR2	Input text containing a valid email address: "send it to test@sliit.lk".	The system correctly identifies "test@sliit.lk" as an EMAIL entity.
TC-PII-03	FR3	Input text containing a common name: "I met with John Doe today".	The system correctly identifies "John Doe" as a PERSON entity.
TC-PII-04	FR2, FR3	Input text with no PII: "This is a great project."	The system returns no detected entities. This is a negative test case.
TC-OS-01	FR4	Input text indicating future travel: "Can't wait for our trip to Kandy next month".	The system identifies the LOCATION entity and flags a TRAVEL_EXPOSURE OpSec risk.
TC-OS-02	FR4	Input text with a location but no temporal risk: "I once visited Kandy."	The system identifies the LOCATION entity but does not flag an OpSec risk.
TC-UI-01	FR6, FR7	A risk is detected in a piece of text.	A visual alert icon appears next to the text. Hovering over the icon displays a tooltip.

8.6 Wireframes

The user interface for the browser extension alerts is designed to be minimal and non-intrusive. The wireframes in **Appendix B** illustrate the intended design, showing the appearance of the subtle alert icon and the on-hover tooltip that provides the educational explanation to the user.

9. PROJECT PLAN AND WORK BREAKDOWN STRUCTURE (WBS)

This section outlines the detailed plan for executing the project, from initial research to final delivery. The project is divided into five distinct phases. The following Work Breakdown Structure (WBS) details the key tasks within each phase, assigns responsibilities to team members, and provides estimated durations. A detailed timeline is visualized in the Gantt Chart located in **Appendix A**.

(Note: M1 = Herath (PII/OpSec, Lead), M2 = Member 2 (Hate Speech), M3 = Member 3 (Misinformation), M4 = Member 4 (Scam, UI))

Phase 1: Research and Foundation (Duration: 4 Weeks)

Table 9.1: Phase 1 of the Work Breakdown Structure: Research and Foundation

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
1.1	Comprehensive Literature Survey for all modules	All	Finalized Reference List, Annotated Bibliography	2 Weeks
1.2	Technology Stack Setup & Environment Configuration	M1	Shared GitHub repository, Configured dev environments	1 Week
1.3	Finalize System Architecture and API "Contract"	M1	Final System Diagrams, API Specification Document	1 Week
1.4	Initial Data Sourcing and Feasibility Assessment	All	Report on available datasets for each module	1 Week

Phase 2: Core Module Development (Duration: 6 Weeks - Parallel Work)

Table 9.2: Phase 2 of the Work Breakdown Structure: Core Module Development

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
2.1	PII/OpSec: Data Synthesis & Annotation	M1	Annotated training and test dataset (JSONL format)	2 Weeks
2.2	PII/OpSec: Model Development & Implementation	M1	Python script for the hybrid NER/Regex model	3 Weeks
2.3	PII/OpSec: Rule Engine Implementation	M1	Python script for the OpSec contextual rule engine	1 Week
2.4	Hate Speech: Model Fine-tuning and Training	M2	Trained hate speech classification model	6 Weeks
2.5	Misinformation: Model Fine-tuning and Training	M3	Trained manipulative language model	6 Weeks
2.6	Scam: Model Fine-tuning and Training	M4	Trained scam detection model	6 Weeks

Phase 3: System Integration and Frontend Development (Duration: 4 Weeks)

Table 9.3: Phase 3 of the Work Breakdown Structure: System Integration and Frontend Development

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
3.1	Backend API Development (FastAPI)	M1, M3	Functional API endpoints for all modules	2 Weeks
3.2	Browser Extension Frontend Development (JS/HTML/CSS)	M4, M2	UI shell of the browser extension	2 Weeks
3.3	Integrate All Modules with the Backend API	M1	Integrated backend service	1 Week
3.4	Connect Browser Extension to Backend API	M4	End-to-end data flow from browser to backend	1 Week

Phase 4: Testing and Evaluation (Duration: 3 Weeks)

Table 9.4: Phase 4 of the Work Breakdown Structure: Testing and Evaluation

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
4.1	PII/OpSec: Unit & Integration Testing	M1	Unit test report, Successful integration test logs	1 Week
4.2	PII/OpSec: Performance Evaluation & Benchmarking	M1	Evaluation report with Precision, Recall, F1-Score	1 Week
4.3	End-to-End System Testing	All	Bug report, System performance metrics	2 Weeks
4.4	Qualitative User Feedback Collection & Analysis	M4	Survey results summary	1 Week

Phase 5: Finalization and Reporting (Duration: 4 Weeks)

Table 9.5: Phase 5 of the Work Breakdown Structure: Finalization and Reporting

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
5.1	Final Project Report Writing	All	Draft report sections	2 Weeks
5.2	Final Report Compilation and Editing	M1	Complete Final Report document	1 Week
5.3	Final Presentation and Viva Preparation	All	Presentation slides, Demonstration script	2 Weeks
5.4	Final Code Submission and Documentation	All	Well-documented codebase on GitHub	1 Week

10. BUDGET AND BUDGET JUSTIFICATION

This project is designed to be completed with no direct financial cost. The development and execution will rely exclusively on open-source software, publicly available academic datasets, and existing university or personal computing infrastructure.

- **Software Costs:** All planned software, including the Python programming language, the FastAPI framework, the Hugging Face Transformers library, and the PyTorch/TensorFlow deep learning frameworks, are available under open-source licenses at no cost.
- **Hardware Costs:** Development and model training will be conducted on personal laptops and standard university-provided computing resources. No specialized or high-performance computing hardware is required for the scope of this project.
- **Data Acquisition Costs:** The datasets required for training and evaluating the models will be sourced from publicly accessible, academic repositories (e.g., Kaggle, arXiv) and through the creation of a synthetic dataset, which does not incur any cost.

Therefore, a budget of zero is proposed for this research project.

11. COMMERCIALIZATION AND ENTREPRENEURSHIP POTENTIAL

"The Guardian" is designed with significant commercialization potential, positioned as a proactive, educational, and user-centric defense system that addresses a clear gap in the personal cybersecurity market.

11.1 Market Opportunity

With the number of global social media users projected to approach six billion by 2027 [14], the addressable market for tools that enhance user privacy and operational security is vast and growing. This user base represents a significant segment of the global information security market, which continues to see robust growth in spending year-over-year [15]. Current solutions (antivirus, ad-blockers) primarily focus on technical threats but largely neglect the human-layer vulnerabilities that "The Guardian" is designed to mitigate.

11.2 Commercialization Pathways

The proposed freemium and enterprise licensing models align with evolving cybersecurity business strategies, which increasingly focus on subscription-based services and proactive, user-centric solutions to build digital trust, as highlighted in industry reports [16].

- **Freemium Browser Extension (B2C):**
 - **Free Tier:** Provides core protection for explicit PII detection, establishing a large user base and demonstrating value.
 - **Premium Tier:** A subscription (\$2-5/month) unlocks the full suite of modules, including advanced OpSec analysis and customizable alert settings, generating recurring revenue.
- **Enterprise & Institutional Licensing (B2B):**
 - **Corporate Security Training:** License the system to businesses as a continuous training tool, directly addressing the challenge of managing security behavior in organizations [17].

- **Educational Integration:** Partner with schools and universities to integrate "The Guardian" into digital literacy curricula.
- **API & Module Licensing (Technology Partnerships):**
 - **Cybersecurity Firms:** License the specialized detection modules as APIs to established security companies to enhance their existing product suites.

11.3 Value Proposition

The core value of "The Guardian" is rooted in the principles of usable security, aiming to transform the user from the proverbial 'weakest link' into an empowered and educated defender [18].

- **Enhanced Security Awareness:** Empowers users to recognize and prevent risky online behaviors.
- **Non-Intrusive Guidance:** Delivers effective security "nudges" without censorship, preserving user autonomy.
- **Scalability:** The modular design allows for the rapid development and integration of new modules.

11.4 Entrepreneurial Impact

"The Guardian" has the clear potential to evolve beyond an academic project into a sustainable cybersecurity venture. Its innovative blend of AI, psychology, and usable security principles makes it an attractive candidate for venture funding, technology incubator programs, and cybersecurity accelerators. The project addresses a rapidly growing global demand for human-layer security solutions, positioning it for significant entrepreneurial success.

REFERENCES

- [1] I. Ghafir, J. Saleem, M. Hammoudeh, H. Faour, V. Prenosil, and S. Jaf, "Security threats to critical infrastructure: the human factor," *The Journal of Supercomputing*, vol. 74, no. 10, pp. 4986–5002, 2018.
- [2] M. Fire, R. Goldschmidt, and Y. Elovici, "Online social networks: Threats and solutions," *IEEE Communications Surveys & Tutorials*, vol. 16, no. 4, pp. 2019–2036, 2014.
- [3] A. Gruzd and Á. Hernández-García, "Privacy Concerns and Self-Disclosure in Private and Public Uses of Social Media," *Cyberpsychology, Behavior, and Social Networking*, vol. 21, no. 1, pp. 24-32, Jan. 2018.
- [4] D. Van Puyvelde and F. T. Rienzi, "The rise of open-source intelligence," **European Journal of International Security**, Jan. 2025.
- [5] Y. Karimi, A. Squicciarini, and S. Wilson, "Automated Detection of Doxing on Twitter," *arXiv preprint arXiv:2209.08862*, 2022.
- [6] B. A. S. Al-Rimy, M. A. Maarof, and S. Z. M. Shaid, "Ransomware threat success factors, taxonomy, and countermeasures: A survey and research directions," *Computers & Security*, vol. 74, pp. 144–166, 2018.
- [7] F. Stubbs and Ö. Uzuner, "De-identification of psychiatric intake records: Overview of 2016 CEGS N-GRID shared tasks Track 1," *Journal of Biomedical Informatic.*, vol. 75, pp. S4–S18, Jul. 2017.
- [8] G. Gambarelli, A. Gangemi, and R. Tripodi, "Is Your Model Sensitive? SPeDaC: A New Benchmark for Detecting and Classifying Sensitive Personal Data," *arXiv preprint arXiv:2212.01520*, 2022.
- [9] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [10] Z. Akhtar et al., "Named Entity Recognition Tools and Techniques in Custom Natural Language Processing Models," in *2024 International Conference on Business Analytics for Technology and Security (ICBATS)*, Dubai, United Arab Emirates, 2024, pp. 1-6.
- [11] S. Selvam and A. Ghosh, "PANORAMA: A Synthetic PII-Laced Dataset for

Sensitive Data Memorization Analysis," *arXiv preprint arXiv:2403.02102*, 2024.

[12] L. Matsakis, "Do You Have 'Security Fatigue'? Too Many Warnings Makes Us Give Up Trying to Be Safe Online," *WIRED*, Oct. 12, 2016. [Online].

Available: <https://www.wired.com/2016/10/security-fatigue-warnings-makes-us-give-trying-safe-online/>. [Accessed: Aug. 15, 2025].

[13] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.

[14] Statista Research Department, "Number of social media users worldwide from 2017 to 2027," **Statista**, Mar. 2024. [Online]. Available:

<https://www.statista.com/statistics/278414/number-of-worldwide-social-network-users/>. Accessed: Aug. 15, 2025.

[15] Gartner, Inc., "Gartner Forecasts Worldwide Security and Risk Management Spending to Grow 14% in 2024," *Press Release*, Dec. 2023. [Online]. Available:

<https://softwarestrategiesblog.com/2024/03/10/gartner-predicts-solid-growth-for-information-security-reaching-287-billion-by-2027/>. Accessed: Aug. 15, 2025.

[16] PwC, "2024 Global Digital Trust Insights," *PricewaterhouseCoopers*, 2023.

[Online]. Available: <https://www.pwc.com/gx/en/issues/cybersecurity/digital-trust-insights.html>. Accessed: Aug. 15, 2025.

[17] A. Beaument, M. A. Sasse, and M. Wonham, "The compliance budget: managing security behaviour in organisations," in *Proc. New Security Paradigms Workshop*, 2008, pp. 47–58.

[18] M. A. Sasse, S. Brostoff, and D. Weirich, "Transforming the 'weakest link'—a human/computer interaction approach to usable and effective security," *BT Technology Journal*, vol. 19, no. 3, pp. 122–131, 2001.

APPENDICES

Appendix A: Project Gantt Chart

PROJECT TITLE: The Guardian

Group ID: 25-26J-041

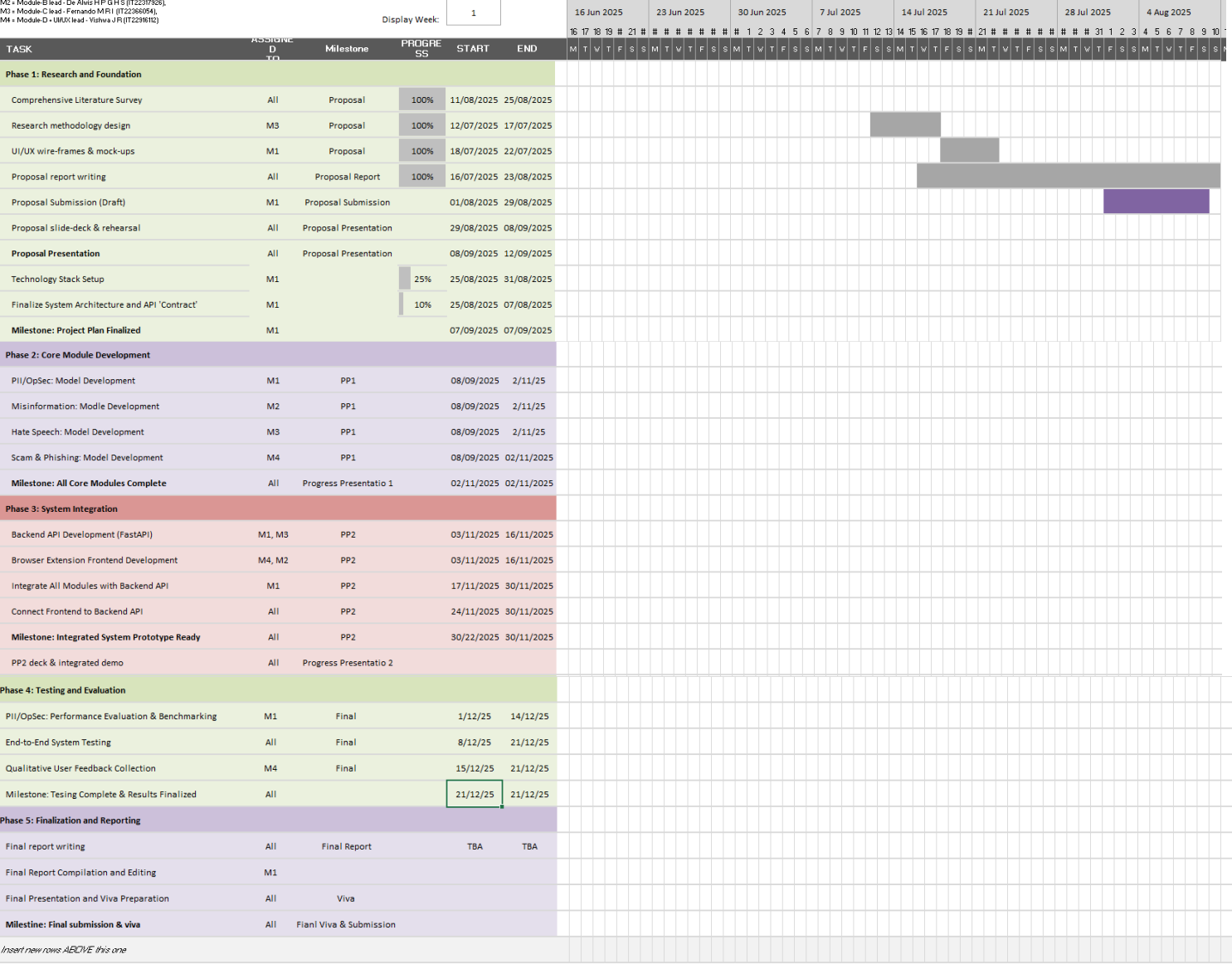
• **Does not**
Members: M1 = Group Group/Module-A lead -Herath H M G D K (IT2236186),
M2 = Module-B lead - De Alwis H P G H S (IT22317826),
M3 = Module-C lead - Fernando M P I (IT22368994),
M4 = Module-D = UMUK lead - Vishwa U R (IT22396102)

Project Start:

Mon, 16/06/2025

Display Week:

1



Appendix B: UI Wireframes



Fig. B.1. Wireframe of the initial alert state. The subtle shield icon appears next to the flagged text without disrupting the user's reading experience, serving as a non-intrusive nudge.

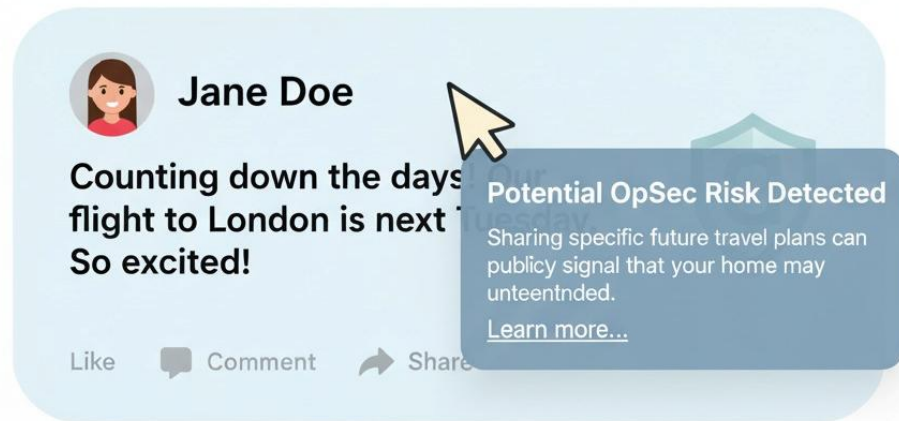


Fig. B.2. Wireframe of the hover state. When the user interacts with the alert icon, a tooltip provides a concise, educational explanation of the specific risk, empowering the user to make an informed decision.